DATA DASH

Presented by:

DATA BASHERS

ELECTIONS DECODED

India's 72% Turnout Average Masks a 36-Point Crisis: A Data-Driven Imperative to Reimagine Democratic Participation

EXECUTIVE SUMMARY

Three Structural Failures Drive India's Participation Crisis, All Three Are Measurable, Targetable, and Fixable

"28% of all assembly elections between 2009–2021 were decided by margins under 5%, making every lost voter a potential lost constituency"

SCALE SUPPRESSION

-0.49

- Correlation: Electorate Size → Turnout
- Every 1 lakh additional voters reduces **turnout by ~2.8 percentage points**. Constituencies above 5 lakh voters average just 52% participation vs. 80% for sub-1 lakh seats.

FRAGMENTATION DRAG

-0.38

- Correlation: More Candidates → Lower Participation
- Body: Constituencies with 20+ candidates average 63% turnout vs. 81% for those with under 5. More choices, paradoxically, suppress democratic participation.

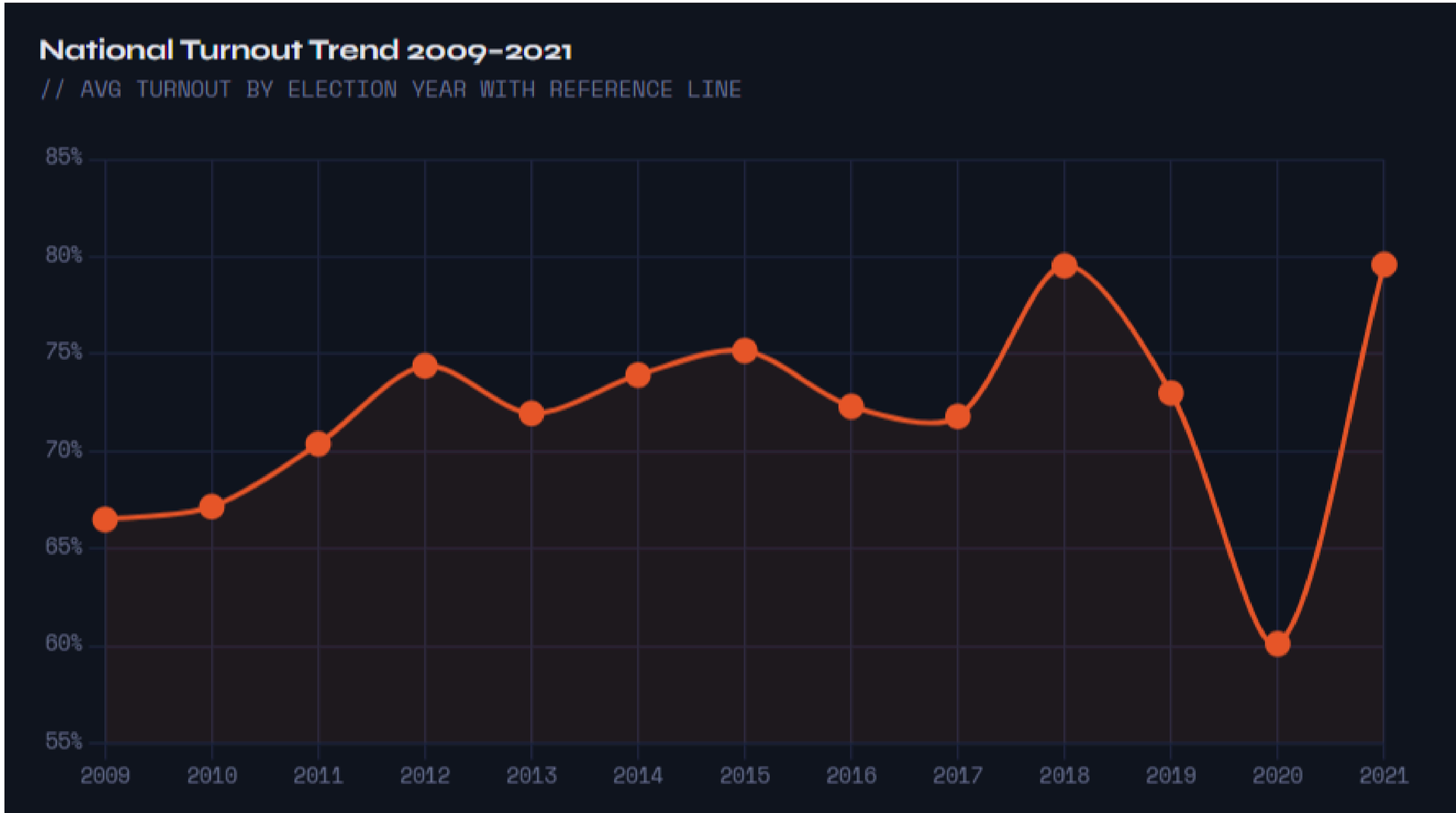
URBAN APATHY

<40%

- Turnout in Urban Bihar & Maharashtra Seats
- Body: Patna's Bankipur, Kumhrarh, Mumbai's Colaba and Versova have delivered sub-40% turnout across 3+ consecutive election cycles.

National Turnout Rose from 67.5% to 78% Over 12 Years, But the Story Beneath the Average Tells a Very Different Truth

32 States & UTs Analysed
4,041 Unique Constituencies
36 pts Turnout Gap (Tripura 92% → Bihar 56%)
1,734 Unique Parties Contested

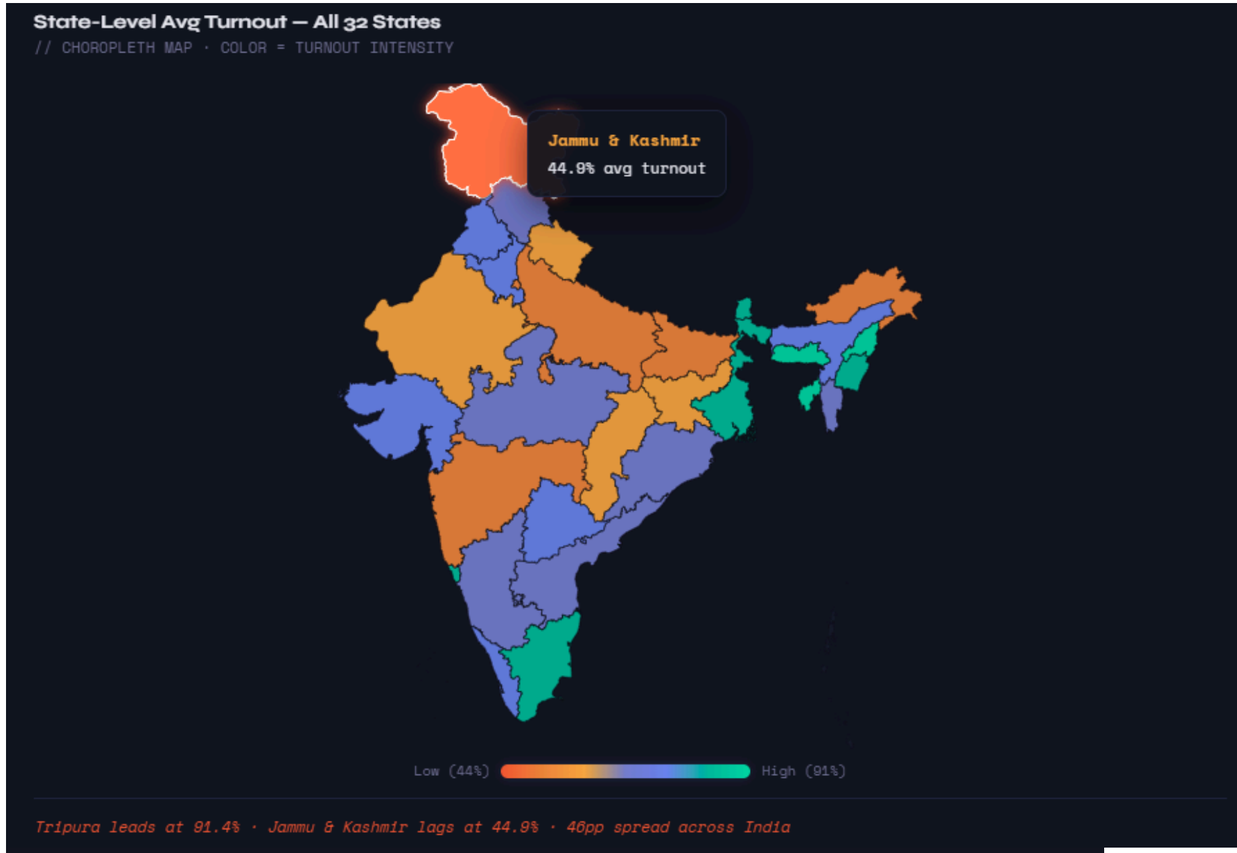


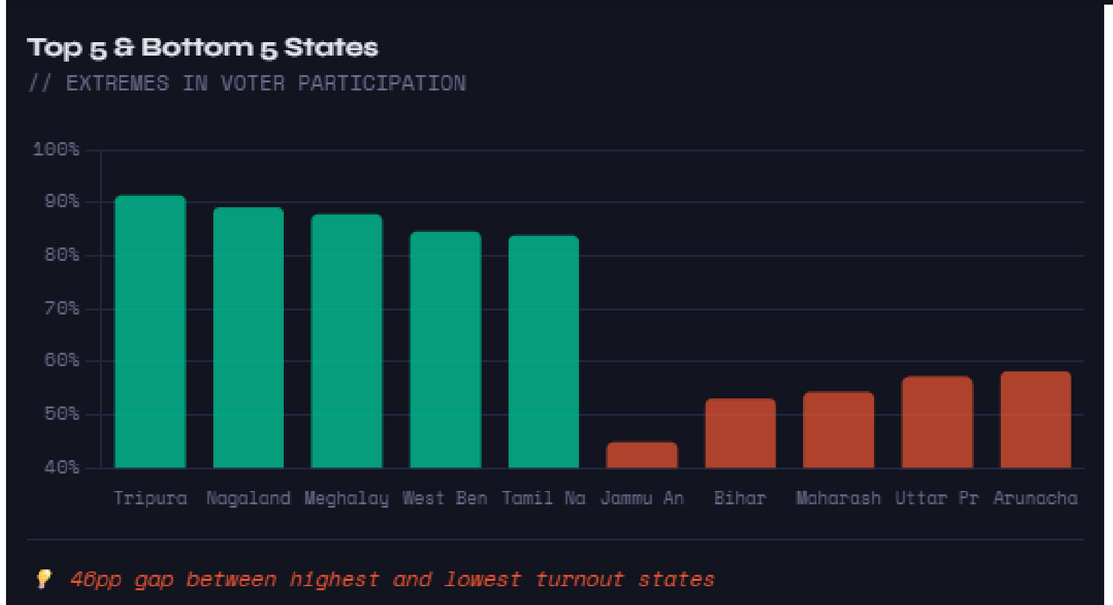
The Participation Divide

India's Participation Crisis is Geographically Concentrated: The BIMARU Belt Drags the National Average by ~15 Percentage Points

Bihar + UP + Maharashtra alone account for ~35% of India's total electorate: fixing these three states is a national priority

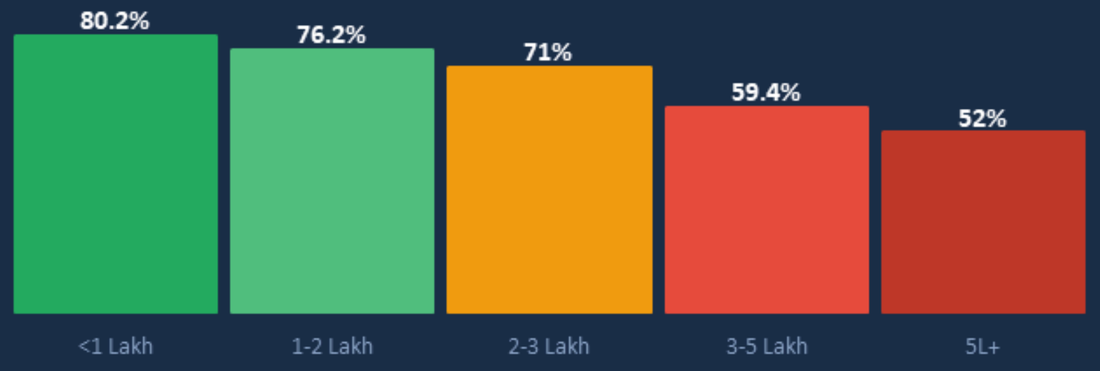
- 🔴 Red states (below 65%): Bihar 56.1%, UP 60.6%, Maharashtra 61.8%
- 🟡 Yellow (65–75%): Delhi, Gujarat, Haryana, Karnataka
- 🟢 Green (above 75%): Tripura 92.4%, Nagaland 88.3%, West Bengal 83%, Goa 82.4%





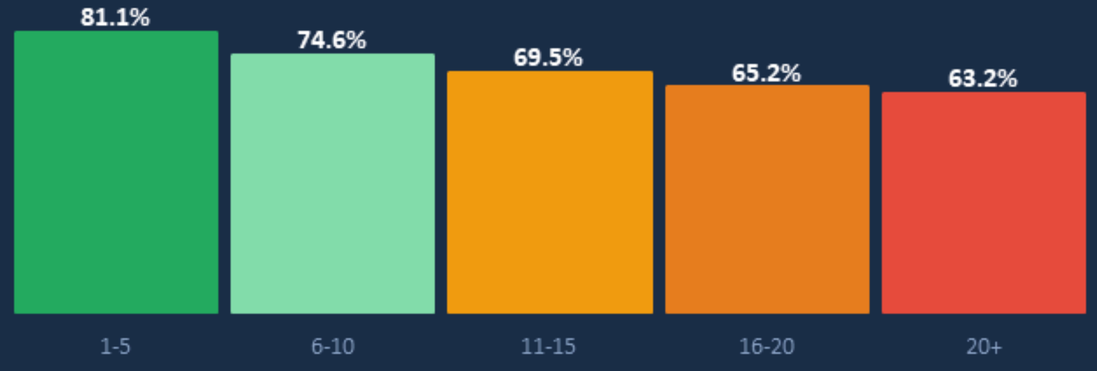
Electorate Size is the Single Strongest Predictor of Low Turnout: A Structural Flaw Election Authorities Can Directly Address

Avg Turnout by Electorate Size



r = -0.49 | Strongest structural predictor in dataset

Avg Turnout by No. of Candidates



r = -0.38 | Each bracket of 5 more candidates = ~4.5% turnout drop

Avg Turnout by Reservation Type

ST constituencies outperform GEN by 6.5 pts — community mobilisation effect

Contest Closeness & Participation

67.1%
Ultra-Close Contests
(<2% margin)

VS

74.3%
Landslide Contests
(25%+ margin)

Perceived foregone conclusions suppress voter motivation | 28% of contests decided by <5% margin

Source: Assembly Elections Dataset 2009–2021 | 32 States | 10,572 Constituency-Year Observations | Pearson Correlation Coefficients

Competitiveness & Electoral Dynamics

Nearly 1-in-3 Assembly Elections Were Decided by Razor-Thin Margins - Voter Apathy in Low-Turnout Seats is Literally Changing Electoral Outcomes

2,990 elections (28%) won by <5% votes - one polling booth per constituency could have changed the result

Party	2009 Wins	2021 Wins	Trend
BJP	312	848	↑ 172%
INC	618	180	↓ 71%
AITC	48	213	↑ 344%
Regional	412	509	↑ 24%
IND	38	52	↑ 37%

Bipolar national contest has fragmented into regional strongholds: turnout patterns diverge accordingly

- **246 Independent winners across 12 years**
- **91.2% Male winner rate - gender gap persists**
- **51.5 yrs Average age of winning candidates**

The Gender Participation Gap

WOMEN COMPRISE 50% OF INDIA'S ELECTORATE BUT ONLY 8.8% OF ITS WINNERS: THE REPRESENTATION DEFICIT IS THE INVISIBLE BARRIER TO TRUE DEMOCRATIC PARTICIPATION

The Gender Funnel

- All Candidates: ~100% are eligible → 7.9% are women (9,469 of 122,559)
- Contested: 7.9% women candidates
- Winners: 8.8% women winners

"Women win at a slightly higher rate than their candidacy share (8.8% wins vs 7.9% nominations): the barrier is candidacy, not electability"

Best performing (women winner %):
- Andhra Pradesh: ~10.4%
- Manipur: ~10.1%
- West Bengal: ~10%

Worst performing:
- J&K, Bihar, UP: <5% each

Key Gender Insights

// STRUCTURAL ANALYSIS

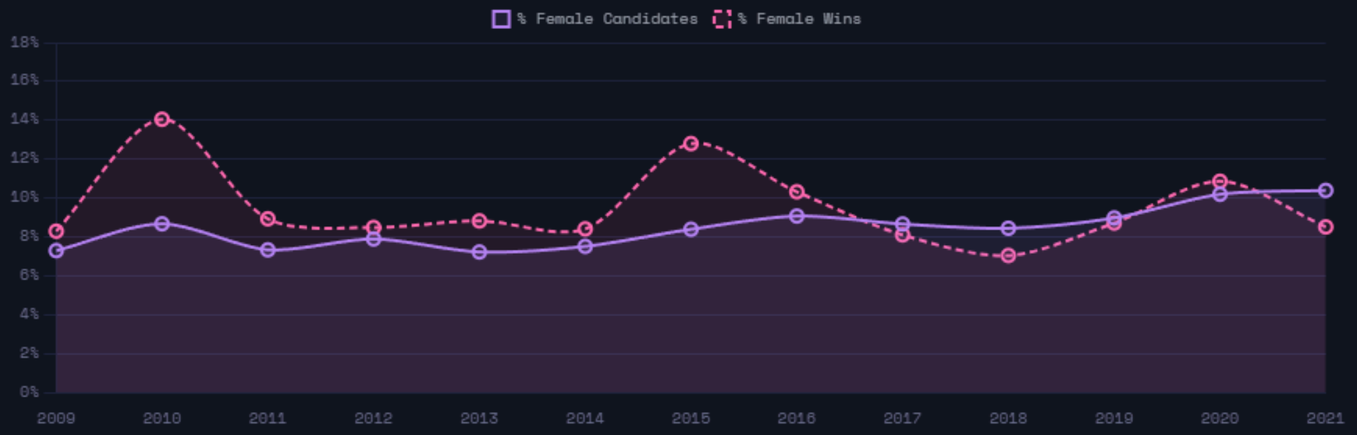
10.4%
Female candidates in 2021 — highest in dataset

11.4%
Peak female winner % — Andhra Pradesh 2009

13yr
Stagnation — only 3pp gain in female candidacy ove

Female Candidate % vs Win Rate — National Trend

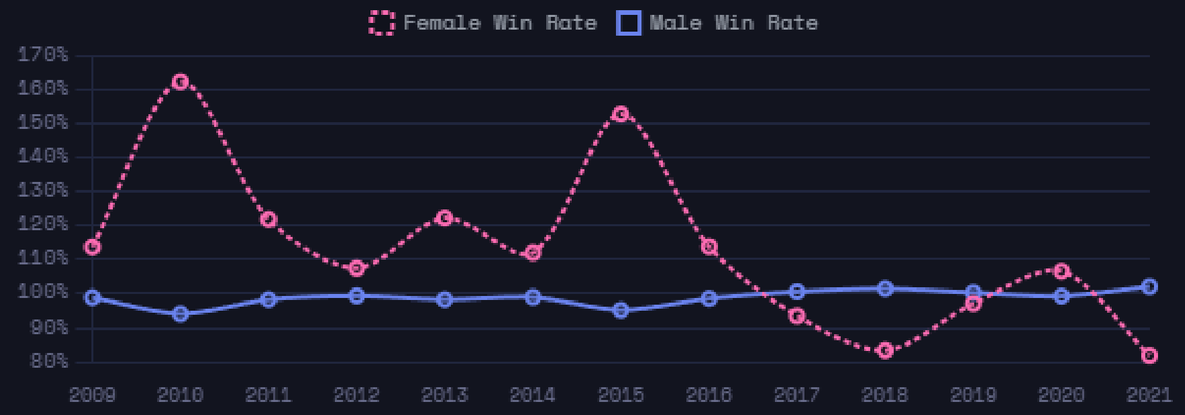
// 2009-2021 · FEMALE SHARE OF CANDIDATES AND WINS



💡 *Female candidates: 7.3% → 10.4% · Female wins: roughly parallel — small but consistent gap*

Gender Win Rate Comparison

// WIN RATE: FEMALE VS MALE CANDIDATES



Not All Low-Turnout Constituencies Are the Same: A Four-Dimensional Priority Index Identifies Where Intervention Delivers Maximum Democratic Impact

Constituency Priority Index (CPI)

Composite score combining 4 structural indicators

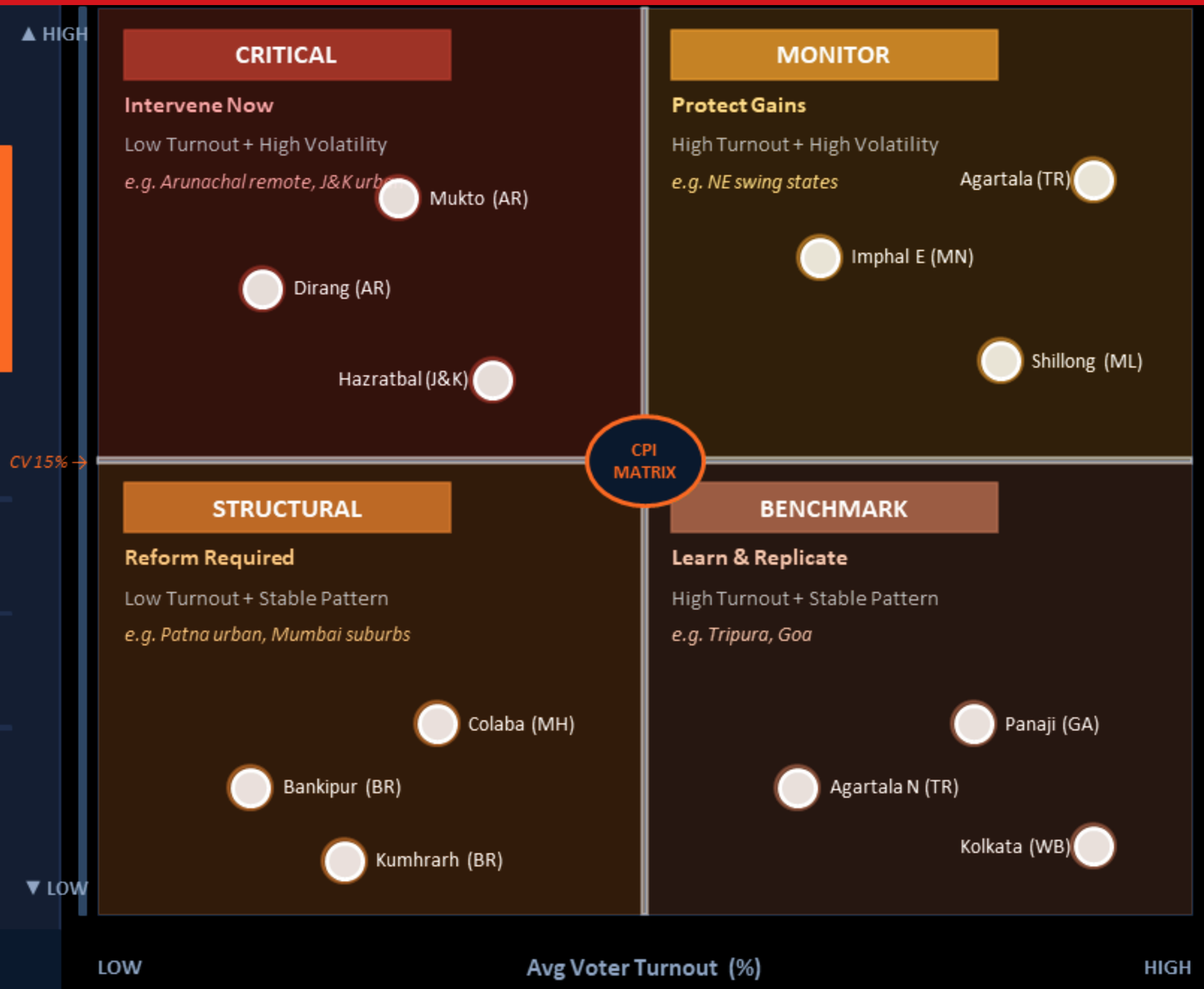
```
CPI  =  0.35 × Low Turnout Score
      + 0.25 × Volatility Score (CV)
      + 0.20 × Electorate Size Score
      + 0.20 × Declining Trend Score
```

35% **Low Turnout Score**
How far below national avg — highest impact lever

25% **Volatility Score**
CV of turnout across cycles — unstable = improvable

20% **Electorate Score**
Larger constituency = bigger impact if recovered

20% **Decline Score**
Falling trajectory signals urgent need



OUTPUT · CPI score 0–1 | Tiered into 4 intervention categories

Source: Assembly Elections Dataset 2009–2021 | CPI = Constituency Priority Index | CV = Coefficient of Variation of Turnout across election cycles

Priority Intervention Targets

20 Constituencies Across 8 States Require Immediate Targeted Intervention: Recovering These Seats to National Average Would Add 18 Million Votes to India's Democratic Mandate

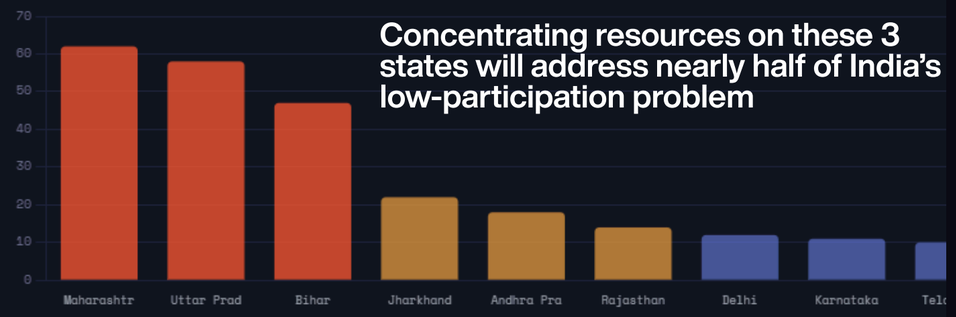
The Impact

If These 50 Seats Reached National Average (72%):

- Current avg turnout in priority seats: ~41%
- Gap: +31 percentage points
- Avg electorate per seat: ~380,000
- Estimated additional votes: ~5.9 million votes per election cycle
- Across 50 seats × typical 3 election cycles: ~18 million reclaimed votes

High Priority States — Constituency Count

// STATES WITH MOST HIGH/CRITICAL PRIORITY CONSTITUENCIES



Concentrating resources on these 3 states will address nearly half of India's low-participation problem

Maharashtra, UP and Bihar account for 40%+ of all High Priority constituencies

Concentrating resources on these 3 states will address nearly half of India's low-participation problem

CPI Methodology: Composite score from (1) Low Turnout Score, (2) Volatility Score, (3) Electorate Size Score, (4) Decline Score. Higher = more urgen
(CPI = 0.35 × Low_Turnout_Score + 0.25 × Volatility_Score + 0.20 × Electorate_Size_Score + 0.20 × Decline_Score)

Top 20 Highest-Priority Constituencies

// RANKED BY CPI SCORE · INTERVENTION TARGETS

RANK	CONSTITUENCY	STATE	AVG TURNOUT	CV	CPI SCORE	TIER	TREND
1	Dirang	Arunachal Pradesh	28.4%	173.2	0.7243	Critical	↓ Sharply Declining
2	Mukto	Arunachal Pradesh	20.7%	200.0	0.6084	Critical	↑ Sharply Improving
3	Sahibabad	Uttar Pradesh	49.3%	0.2	0.5160	Critical	→ Stable
4	Along East	Arunachal Pradesh	51.2%	86.8	0.4981	High	↓ Sharply Declining
5	Digha	Bihar	39.1%	6.9	0.4793	High	→ Stable
6	Kumhrarh	Bihar	36.9%	4.1	0.4787	High	→ Stable
7	Northern Angami-II	Nagaland	60.6%	86.8	0.4784	High	↓ Sharply Declining
8	Yachuli	Arunachal Pradesh	58.1%	87.0	0.4768	High	↓ Sharply Declining
9	Hazratbal	J&K	24.9%	26.2	0.4742	High	↑ Improving
10	Noida	Uttar Pradesh	45.3%	13.1	0.4697	High	→ Stable

Arunachal Pradesh and Bihar dominate the critical/high-priority list — urban UP metros like Noida and Lucknow also flagged

Urban apathy in metros is as dangerous as rural remoteness — both require fundamentally different intervention strategies

Three Evidence-Based Interventions Can Recover 6-18 Million Lost Votes: Start With Urban Constituencies, Deliver ROI Within One Election Cycle

01 — STRUCTURAL REBALANCING

Constituency Delimitation Reform

TARGET	All constituencies >3L voters in Bihar, UP, Maharashtra
ACTION	Sub-delimitation of mega-constituencies (500k+ voters) into manageable units
EVIDENCE	Constituencies <1L → 80.2% turnout vs 52% for 5L+ seats
IMPACT	+6–10% turnout lift in affected constituencies

ROI: Very High | Medium (legislative) | 12–24 months

02 — TARGETED ENGAGEMENT

Urban Constituency Mobilisation Programme

TARGET	Top 50 CPI priority constituencies — Bihar + Maharashtra urban
ACTION	Booth-level registration drives, transport facilitation, employer voting-day partnerships
EVIDENCE	Urban Bihar seats avg 37–39% — structural, not attitudinal
IMPACT	+8–15% turnout in ultra-low seats (37% → 45–52%)
COST	₹15–25 Cr / election cycle

ROI: Immediate | High (operational) | 3–6 months pre-election

03 — EARLY WARNING SYSTEM

Predictive Participation Monitoring Dashboard

TARGET	National Electoral Authority — this deliverable
ACTION	Live CPI dashboard tracking 4 KPIs: electorate growth, candidate count, turnout slope, margin trend
EVIDENCE	15+ constituencies show declining trajectory across 3+ cycles — identifiable 2 yrs ahead
IMPACT	Proactive intervention vs. reactive response — ROI multiplied every cycle
COST	₹2–5 Cr build + ₹50L/yr

ROI: Multiplied | Very High | Immediate (6 months)

IMPLEMENTATION ROADMAP

1. **NOW → 3M** — *0–3 months*
   Deploy CPI dashboard · Identify top 50 priority seats
2. **3M → 9M** — *3–9 months*
   Launch urban mobilisation in Bihar, UP, Maharashtra
3. **9M → 18M** — *9–18 months*
   Begin delimitation advocacy for 5L+ constituencies
4. **18M → 36M** — *18–36 months*
   Measure lift · Iterate CPI model · Expand programme

"Democracy's strength is measured not by who wins, but by how many participate. The data shows where we are failing — and exactly how to fix it."

DATA DASH

Presented by:

DATA BASHERS

THANK YOU

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